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Mathematics

ENGLISH

10 Standard

Part –2

Version 1.1

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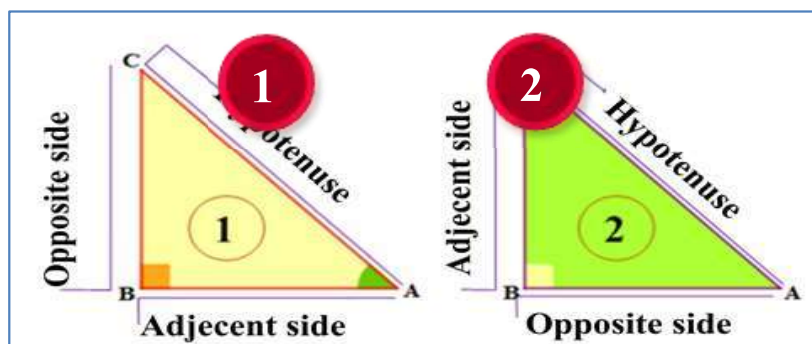
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INTRODUCTION TO
TRIGONOMETRY

Trigonometry is the study of relationships between the sides and angles of a triangle.

11.2 Trigonometric Ratios

To know the trigonometric ratio we have to consider right angle triangle.



Trigonometric ratios	1	2
$\sin A$	$\frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{BC}{AC}$	$\frac{AB}{AC}$
$\cos A$	$\frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{AB}{AC}$	$\frac{BC}{AB}$
$\tan A$	$\frac{\text{Opposite}}{\text{Adjacent}} = \frac{BC}{AB}$	$\frac{AB}{BC}$
$\operatorname{cosec} A$	$\frac{\text{Hypotenuse}}{\text{Opposite}} = \frac{AC}{BC}$	$\frac{AC}{AB}$
$\sec A$	$\frac{\text{Hypotenuse}}{\text{Adjacent}} = \frac{AC}{AB}$	$\frac{AC}{BC}$
$\cot A$	$\frac{\text{Adjacent}}{\text{Opposite}} = \frac{AB}{BC}$	$\frac{BC}{AB}$

There are six trigonometric ratios:

Example 1: $\tan A = \frac{4}{3}$ find the other trigonometric ratios of the angle A

In $\triangle ABC$, $\angle ABC = 90^\circ$

$\therefore$ By Pythagoras theorem,

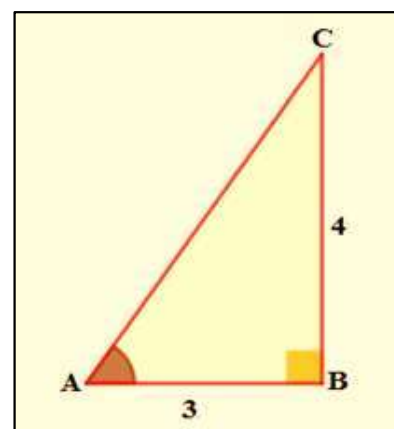
$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow AC^2 = 4^2 + 3^2 = 16 + 9 = 25$$

$$\Rightarrow AC = 5$$

$$\sin A = \frac{BC}{AC} = \frac{4}{5}; \cos A = \frac{AB}{AC} = \frac{3}{5}; \tan A = \frac{BC}{AB} = \frac{4}{3}$$

$$\operatorname{cosec} A = \frac{AC}{BC} = \frac{5}{4}; \sec A = \frac{AC}{AB} = \frac{5}{3}; \cot A = \frac{AB}{BC} = \frac{3}{4}$$



Example 2: If B and Q are acute angles such that $\sin B = \sin Q$, then prove that $\angle B = \angle Q$.

$$\sin B = \sin Q$$

$$\Rightarrow \frac{AC}{AB} = \frac{PR}{PQ}$$

$$\Rightarrow \frac{AC}{PR} = \frac{AB}{PQ} = k \quad \text{----- (1)}$$

$$BC = \sqrt{AB^2 - AC^2} \quad [\text{By Pythagoras theorem}]$$

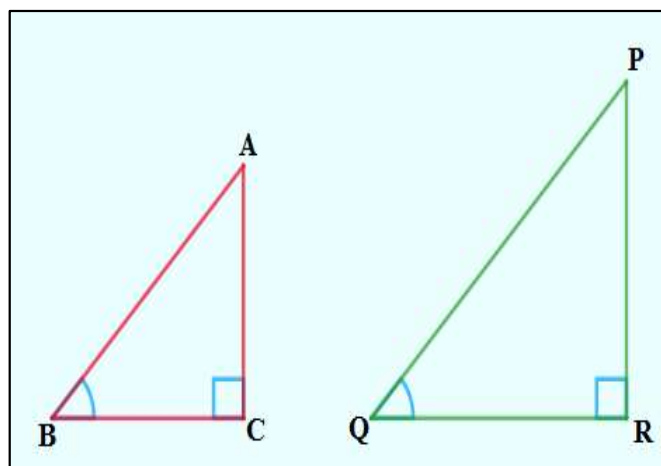
$$\Rightarrow \sqrt{k^2 PQ^2 - k^2 PR^2}$$

$$\Rightarrow k \cdot \sqrt{PQ^2 - PR^2} \quad [\text{from (1)}]$$

$$QR = \sqrt{PQ^2 - PR^2}$$

$$\Rightarrow \frac{BC}{QR} = \frac{k \cdot \sqrt{PQ^2 - PR^2}}{\sqrt{PQ^2 - PR^2}} = k \quad \text{----- (2)}$$

From (1) and (2),



$$\frac{AC}{PR} = \frac{AB}{PQ} = \frac{BC}{QR}$$

$$\Rightarrow \triangle ABC \sim \triangle PQR$$

$$\therefore \angle B = \angle Q$$

Example 3 : Consider $\triangle ACB$, right-angled at C, in which $AB = 29$ units, $BC = 21$ units and $\angle ABC = \theta$ (see Fig. 8.10). Determine the values of $\cos^2 \theta + \sin^2 \theta$ (ii) $\cos^2 \theta - \sin^2 \theta$

In right angle triangle ACB , $\angle ACB = 90^\circ$

$$\text{Therefore } AC = \sqrt{AB^2 - BC^2}$$

$$\Rightarrow AC = \sqrt{29^2 - 21^2}$$

$$\Rightarrow AC = \sqrt{841 - 441} = \sqrt{400} = 20$$

$$(i) \quad \cos^2 \theta + \sin^2 \theta = \frac{21^2}{29^2} + \frac{20^2}{29^2} = \frac{441+400}{841} = \frac{841}{841} = 1$$

$$(ii) \quad \cos^2 \theta - \sin^2 \theta = \frac{21^2}{29^2} - \frac{20^2}{29^2} = \frac{441-400}{841} = \frac{41}{841} = 1$$

Example 4 : In a right triangle ABC , right-angled at B, if $\tan A = 1$, then verify that $2 \sin A \cos A = 1$

In right angle triangle ACB ,

$$\tan A = 1 \Rightarrow \frac{AB}{BC} = 1$$

$$\Rightarrow AB = BC$$

$$AC^2 = AB^2 + BC^2 \text{ [By Pythagoras theorem]}$$

$$\Rightarrow AC^2 = 2AB^2 \text{ ----- (1)}$$

$$\text{Now, } 2 \sin A \cos A = 2 \cdot \frac{AB}{AC} \cdot \frac{BC}{AC}$$

$$= 2 \cdot \frac{AB^2}{AC^2}$$

$$= 2 \cdot \frac{AB^2}{2AB^2} = 1$$

Example 5 : In $\triangle OPQ$ right-angled at P, $OP = 7$ cm and $OQ - PQ = 1$ cm (see Fig. 11.12). Determine the values of $\sin Q$ and $\cos Q$.

In $\triangle OPQ$, $OQ^2 = PQ^2 + OP^2$ [By Pythagoras theorem]

$$\Rightarrow (1 + PQ)^2 = PQ^2 + 7^2$$

$$\Rightarrow 1 + PQ^2 + 2PQ = PQ^2 + 49$$

$$\Rightarrow 1 + 2PQ = 49$$

$$\Rightarrow 2PQ = 49 - 1 = 48$$

$$\Rightarrow PQ = 24 \text{ cm}$$

$$\Rightarrow OQ = 1 + PQ = 1 + 24$$

$$\Rightarrow OQ = 25$$

$$\therefore \sin Q = \frac{7}{25} \text{ and } \cos Q = \frac{24}{25}$$

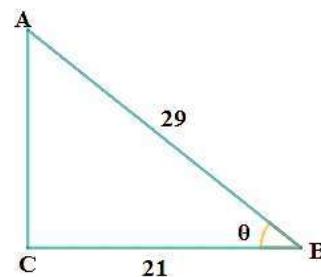


Fig. 8.10

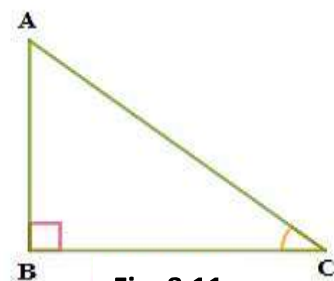


Fig. 8.11

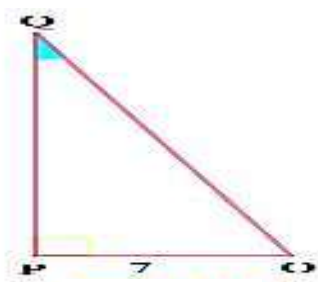


Fig. 8.12

Inverse of trigonometric values

$\frac{1}{\sin A}$	$\frac{\text{Hypotenuse}}{\text{Opposite}}$	CosecA
$\frac{1}{\cos A}$	$\frac{\text{Hypotenuse}}{\text{Adjacent}}$	SecA
$\frac{1}{\tan A}$	$\frac{\text{Adjacent}}{\text{Opposite}}$	CotA

Inverse of trigonometric values

$\frac{1}{\text{CosecA}}$	$\frac{\text{Opposite}}{\text{Hypotenuse}}$	SinA
$\frac{1}{\text{SecA}}$	$\frac{\text{Adjacent}}{\text{Hypotenuse}}$	SecA
$\frac{1}{\text{CotA}}$	$\frac{\text{Opposite}}{\text{Adjacent}}$	CotA